GPU Power Profiles in AI Data Centers

Understanding GPU Power Profiles in AI Data Centers

Motivation

As AI training clusters grow to hundreds of megawatts, understanding their power consumption behavior has become critical for grid planning and data center operations. Choukse et al. (2025) — *Power Stabilization for AI Training Datacenters* (arXiv: 2508.14318, Microsoft / OpenAI / NVIDIA) — present the first detailed characterization of power dynamics in large-scale AI training facilities. Their key findings motivate this project:

GPU power is not constant during training. Each training iteration cycles through compute-heavy phases (forward/backward pass, drawing near-TDP power) and communication phases (gradient synchronization, where GPU power drops significantly). Because bulk-synchronous parallelism keeps thousands of GPUs in lock-step, these cycles are **correlated across the entire cluster**, producing large aggregate power swings rather than averaging out.

These swings create real problems for the electrical grid:

1. **Ramp rate violations.** Rapid power transitions between training phases can exceed the MW/minute ramp limits that utilities impose on large loads. Choukse et al. show that AI data centers can violate ramp rate thresholds much more frequently than conventional workloads.
2. **Dynamic power range and voltage flicker.** The difference between peak and trough power during a training job determines the dynamic power envelope the grid must accommodate. Large swings can cause voltage fluctuations on distribution feeders, known as flicker.
3. **Frequency-domain concerns.** The periodic nature of training iterations injects power oscillations at specific frequencies. Inter-area oscillation modes (roughly 0.2–3 Hz) are a known grid stability concern, and per-iteration cycling of large GPU clusters can fall in or near these bands. Torsional resonance of turbine-generator shafts occurs at higher frequencies (roughly 7–100+ Hz) and is a separate concern; the point is that periodic load patterns at any frequency deserve grid-impact analysis.

Note on time scales: The per-iteration power oscillations documented in Choukse et al. (their Figure 1 and StratoSim simulations) occur at sub-second granularity within individual training steps. Our simulation experiment does *not* capture these intra-

iteration dynamics — we profile at the workload-phase level, measuring the average power envelope of each task type (Pretrain, SFT, MLLM, Eval) over tens of seconds to minutes. This gives us the **static power profile per workload type**, which is the building block for estimating aggregate data center demand when combined with job scheduling traces. Capturing the fast intra-step oscillations would require sub-millisecond instrumentation and is left for future work.

This project takes a first step toward characterizing workload-level power profiles: we analyze real data center job traces to understand workload composition, then measure per-GPU power draw for representative AI tasks on local hardware.

Project Flow

```
---
config:
  look: handDrawn
  theme: neutral
---
flowchart TB
    subgraph DM["Workload Demand Modelling()"]
        direction TB
        A1["AcmeTrace Dataset\n──────────────\nSeren · 
Kalos\nPhilly · Helios · PAI"]
        A2["Job Types\n──────────────\nPretrain · SFT\nMLLM · 
Eval"]
        A3["Job Attributes\n──────────────\nGPU 
count\nDuration\nQueue time"]
        A1 --> A2 --> A3
    end

    subgraph DF["Workload Demand Forecasting"]
        direction TB
        A4["GPU count distribution\nper task type"]
        A5["Cluster utilization\ntimeline"]
        A4 --- A5
    end

    subgraph PM["Power Profile Modelling()"]
        direction TB
        B1["Local Server\n──────────────\n2× RTX 3080\nNVML @ 
50ms"]
        B2["4 Workloads\n──────────────\nPretrain: BERT\nSFT: 
GPT-2\nMLLM: BLIP-2\nEval: GPT-2"]
        B1 --> B2
    end

    subgraph PP["Per-GPU Power Profiles"]
        direction TB
        B3["Pretrain ~332 W/GPU\nSFT      ~330 W/GPU\nMLLM    ~278 
W/GPU\nEval      ~158 W/GPU"]
    end

    subgraph SC["Scaling: DDP Linear Assumption"]
```

```
    direction TB
    C1["Per-GPU power ×\nGPU count per job\n= Job power (kW)"]
end

A3 --> DF
B2 --> PP

DF -->|"GPU count\n& schedule"| SC
PP -->|"W/GPU\nper type"| SC

subgraph GI["⚡ Grid Impact Assessment"]
    direction LR
    G1["Ramp Rate\nMW/min\nviolations"]
    G2["Voltage\nFlicker\nΔV swings"]
    G3["Frequency\nOscillation\n0.2–100+ Hz"]
end

SC -->|"aggregate\ncluster power"| GI

subgraph DATA["Data Sources"]
    direction LR
    DA["AcmeTrace: real traces\n(historical, 2023)"]
    DB["NVML: measured power\n(local experiment)"]
end

DATA -.->|"input"| DM
DATA -.->|"input"| PM
```

Part 1: Data Center Workload Composition

Data Source

We use the **AcmeTrace** dataset (published on HuggingFace: `Qinghao/AcmeTrace`), which contains job-level traces from five production GPU clusters: **Seren, Kalos, Philly, Helios, and PAI**.

After inspecting all five subsets, we found that only **Seren** and **Kalos** include job-type labels. Among them, Seren provides full coverage of the four target types (Pretrain, SFT, MLLM, Eval), while Kalos lacks MLLM jobs (and has only 6 SFT records). The other three clusters (Philly, Helios, PAI) do not record job types at all. Therefore, our analysis primarily relies on the **Seren** trace.

Data Cleaning

From the raw Seren trace (~1 million jobs), we filtered out non-GPU job categories (`Other`, `Debug`, `CPU`), kept only successfully completed jobs (`state == COMPLETED`), and removed zero-duration entries. This yields **179,199 valid GPU jobs** across four types:

Task Type	Job Count	Proportion
Eval	134,269	74.9%
SFT	39,326	21.9%
MLLM	4,656	2.6%
Pretrain	948	0.5%

Key Findings: GPU Usage by Task Type

The following table summarizes the GPU allocation per job across types:

Stat	Eval	MLLM	Pretrain	SFT
Mean GPUs	2.3	5.7	**220.8**	6.1
Median GPUs	1	2	**128**	8
5th pctl	1	1	48	1
95th pctl	8	8	**768**	8
Max GPUs	128	256	**1,024**	512

Pretrain jobs use far more GPUs than any other type (median 128, up to 1,024), while Eval and SFT typically use 1–8 GPUs. This aligns with the industry pattern where large-scale pretraining requires massive data-parallel or model-parallel GPU clusters.

Job Duration by Task Type

Stat	Eval	SFT	MLLM	Pretrain
Mean	0.35 h	0.27 h	**1.71 h**	**1.51 h**
Median	0.09 h	0.09 h	0.18 h	0.02 h
5th pctl	0.03 h	0.02 h	0.01 h	~0 h
95th pctl	1.02 h	0.76 h	**6.98 h**	**8.68 h**
Max	324.11 h	117.54 h	91.00 h	146.64 h

MLLM and Pretrain jobs run much longer on average (hours to days) compared to Eval/SFT (minutes to hours). At the 95th percentile, Pretrain jobs reach ~8.7 hours and MLLM ~7.0 hours.

Queue Time by Task Type

Stat	Eval	SFT	MLLM	Pretrain
Mean	0.07 h	0.05 h	**0.20 h**	0.02 h
Median	~0 h	~0 h	~0 h	~0 h
95th pctl	0.26 h	0.15 h	**0.62 h**	0.01 h
Max	19.03 h	15.91 h	15.45 h	4.35 h

MLLM jobs wait longest (mean 0.20 h); Pretrain jobs (despite high GPU count) have remarkably short queue times (mean 0.02 h), suggesting they receive priority scheduling.

Overall Cluster Utilization

Across the Seren trace (~173-day span, 2023-03-02 to 2023-08-21):

Stat	Concurrent GPUs
Mean	**259**
Median	178
25th pctl	11
75th pctl	332
95th pctl	993
Peak (max)	**1,705**
Idle time (0 GPUs)	21.7%

The cluster runs an average of **~259 GPUs concurrently** (median 178), peaking at **1,705 GPUs** simultaneously. Roughly 21.7% of minutes have zero active GPUs, indicating significant idle periods.

Part 2: GPU Power Profiling via Simulation

Setup

We used our own server equipped with **2 × NVIDIA GeForce RTX 3080 GPUs** to simulate the four workload types. Power consumption was sampled at 50 ms intervals via NVML (`pynvml`), and the experiment was orchestrated using PyTorch's `torchrun` with Distributed Data Parallel (DDP) across both GPUs.

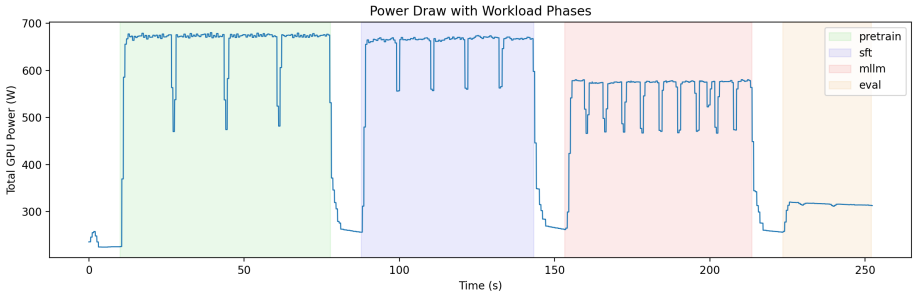
Workloads Simulated

Each workload uses a representative model architecture:

Phase	Model	Description
Pretrain	bert-base-uncased (Transformer encoder)	Full forward + backward pass training
SFT	GPT-2 (CausalLM)	Supervised fine-tuning with gradients
MLLM	BLIP-2 (Salesforce/blip2-opt-2.7b)	Vision-to-text multimodal training
Eval	GPT-2 (Autoregressive decoding)	Inference-only, no gradient computation

The phases ran sequentially with idle gaps between them (10 s each) for clear phase separation.

Results



Power Draw with Workload Phases

Phase	Mean Power (W)	Std (W)	Min (W)	Max (W)
Pretrain	**663.5**	38.9	470.1	680.4
SFT	**658.9**	28.8	556.0	674.2
MLLM	**556.1**	37.3	466.3	580.4
Eval	**315.7**	1.7	311.4	319.4

Total power across 2 GPUs; per-GPU values are approximately half.

Observations

- **Pretrain ≈ SFT >> MLLM >> Eval** in mean power draw. Training workloads (forward + backward + optimizer) are substantially more power-hungry than inference.
- **Eval power is remarkably stable** (std ≈ 1.7 W), reflecting the steady-state nature of autoregressive decoding without gradient computation.
- **Pretrain and MLLM show higher variance** (std ≈ 37–39 W), likely due to intermittent communication overhead and memory-intensive operations.
- SFT is slightly below Pretrain because our toy SFT model (GPT-2) is smaller in parameter count and feedforward width than the Pretrain model; in practice, pretraining usually draws more power than SFT on the same model due to longer sequences and larger batch sizes.

Why Single-Server Profiling Scales to Data Center Estimation

In data center training, a model is trained using **data parallelism** (e.g., DDP) across many GPUs. In this paradigm, each GPU independently processes a shard of the data and performs the same computation — forward pass, backward pass, and gradient synchronization. Because every GPU executes the **same workload in lock-step**, the per-GPU power profile remains consistent regardless of cluster size. The small communication overhead for gradient all-reduce adds minimal power draw beyond the compute portion.

Therefore, the per-GPU power profile measured on our 2-GPU server can be **linearly scaled** by the number of GPUs to estimate total cluster power for a given job. For example: - A Pretrain job using 128 GPUs would draw approximately $128 \times 332 \approx 42.5$ kW (using ~332 W per GPU). - An Eval job using 8 GPUs would draw approximately $8 \times 158 \approx 1.3$ kW.

Combined with Part 1's workload composition data (GPU count distribution and job type proportions), this enables estimation of aggregate power demand across a data center.

Summary

Aspect	Key Takeaway
Job mix	Eval jobs dominate by count (~75%), but Pretrain dominates GPU-hours due to massive GPU allocation (median 128 GPUs)
GPU range	Pretrain: 48–768 GPUs (5th–95th pctl); SFT/MLLM/Eval: typically 1–8 GPUs
Power per GPU	Pretrain/SFT: ~330 W; MLLM: ~278 W; Eval: ~158 W
Scalability	Per-GPU profiles scale linearly due to synchronous data-parallel training
Cluster utilization	Average ~259 GPUs active concurrently, peak ~1,705 GPUs

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References

- Choukse, E. et al. (2025). *Power Stabilization for AI Training Datacenters.* arXiv:2508.14318. Microsoft / OpenAI / NVIDIA.
- AcmeTrace dataset: `Qinghao/AcmeTrace` on HuggingFace.

Repository Structure

```
Power draw/
├── README.md                   # This document
├── requirements.txt            # Python dependencies
├── .gitignore
├── data/                       # AcmeTrace job traces (download
from HuggingFace)
│   ├── cluster_summary.csv
│   ├── trace_seren.csv         # Primary dataset used in analysis
│   ├── trace_kalos.csv
│   ├── helios_trace.csv
│   ├── philly_trace.csv
```

```
│   ├── philly_trace_merge_retry.csv
│   └── pai_trace.csv
├── code/                       # Experiment scripts
│   ├── run.py                  # Main power profiling experiment
(torchrun)
│   ├── LLM_toymodel.py         # Workload definitions (Pretrain/
SFT/MLLM/Eval)
│   └── utils.py                # DDP & NVML utilities
├── notebooks/                  # Analysis notebooks
│   ├── CheckDataset.ipynb      # Part 1: Data center trace
analysis
│   └── result.ipynb            # Part 2: Power profiling results &
visualization
├── results/                    # Experiment outputs (generated by
run.py)
│   ├── timestamps.npy
│   ├── power_per_gpu.npy
│   ├── power_total.npy
│   ├── phase_times.npy
│   ├── phase_labels.npy
│   └── power_draw.png
├── outputs/                    # Rendered HTML/PDF reports
│   ├── CheckDataset.html
│   ├── result.html
│   └── result.pdf
└── reference/                  # Tutorial notebooks (external, for
learning only)
    ├── Pretrain.ipynb
    ├── SFT.ipynb
    ├── MLLM.ipynb
    └── Eval.html
```

How to Reproduce

1. Download AcmeTrace data: `huggingface-cli download Qinghao/AcmeTrace --local-dir data/`
2. Install dependencies: `pip install -r requirements.txt`
3. Run power profiling (requires NVIDIA GPU): `torchrun --standalone --nproc_per_node=2 code/run.py`
4. Open `notebooks/result.ipynb` to visualize results
5. Open `notebooks/CheckDataset.ipynb` to explore the trace data